

A Comparative Analysis of Tree-Based Ensemble Classifiers for Binary Online News Popularity Prediction

Cornelius Christian Setiadi¹⁾, Josua Edward Budianto²⁾, Jovan Melsyah³⁾

¹²³Computer Science & Statistics, Bina Nusantara University

email: cornelius.setiadi@binus.ac.id¹⁾, josua.budianto@binus.ac.id²⁾, jovan.melsyah@binus.ac.id³⁾

Abstract

The rapid growth of digital news platforms has intensified competition for user attention, making the ability to predict article popularity before publication a valuable capability for content strategists and digital publishers. This study proposes a structured machine learning pipeline for binary classification of online news popularity using the Mashable Online News Popularity dataset, comprising 39,644 articles described by 60 predictive attributes. The target variable, raw share count, is transformed into a binary label using the median share count of 1,400 as a classification threshold, producing two classes: Not Popular and Popular. The pipeline incorporates feature decoding to restore interpretability of one-hot encoded categorical variables, a strict split-before-encode strategy to prevent data leakage during categorical encoding, and exploratory data analysis to examine feature distributions across popularity classes. Three tree-based ensemble classifiers are trained and compared: Random Forest, XGBoost, and LightGBM. All models are evaluated in both baseline and hyperparameter-tuned configurations using RandomizedSearchCV with 3-fold cross-validation. Model performance is assessed using accuracy, precision, recall, and F1-score on a held-out test set comprising 20% of the dataset. Experimental results demonstrate that all three tuned models achieve consistent accuracy in the range of 65.68%–67.27%, with tuned LightGBM achieving the highest accuracy of 67.27%, marginally outperforming tuned XGBoost at 67.23% and tuned Random Forest at 65.68%. Feature importance analysis across all three models consistently identifies keyword performance metrics, particularly `kw_avg_avg`, alongside article channel, dominant LDA topic, and self-reference share history as the most influential predictors of article shareability.

Keywords: online news popularity, binary classification, Random Forest, XGBoost, LightGBM, ensemble learning, hyperparameter tuning, RandomizedSearchCV, feature importance, Mashable dataset

1. INTRODUCTION

The exponential growth of digital news platforms has fundamentally reshaped how information is produced, distributed, and consumed in the modern era. Online publishers such as Mashable generate hundreds of articles daily, competing for user attention in an increasingly saturated information landscape. Within this environment, social media shares have emerged as one of the most critical indicators of content success, directly influencing article reach, advertiser interest, and platform revenue. Despite this significance, the factors that determine whether an article will be widely shared remain poorly understood, as online virality is shaped by a complex and often unpredictable combination of content quality, publication timing, topic relevance, and user network behavior [1].

Predicting the popularity of online news articles before publication is therefore of considerable practical value to content strategists, editors, and digital marketing professionals. An accurate predictive model would enable publishers to make informed, data-driven decisions regarding content creation, keyword selection, and publication scheduling, ultimately improving engagement outcomes at scale. However, a fundamental challenge in this prediction task lies in the inherently stochastic nature of human online behavior. Prior work has demonstrated that predicting popularity a priori is difficult precisely because user behavior on social platforms is influenced by probabilistic factors and stochastic evolutions that are difficult to model deterministically. As a result, framing news popularity prediction as a regression problem, attempting to estimate the precise number of shares an article will

receive, proves inherently limited, as the continuous target variable is heavily influenced by unpredictable social dynamics. A more robust and practically meaningful formulation is therefore to reframe the task as binary classification, distinguishing between popular and non-popular articles using the median share count as a balanced threshold, an approach that reduces sensitivity to extreme outliers and yields predictions that are directly actionable for editorial decision-making [2].

In response to these challenges, this study proposes a structured machine learning pipeline for binary online news popularity prediction using the Mashable Online News Popularity dataset, comprising 39,644 articles and 60 predictive attributes. The target variable, raw share count, is transformed into a binary label using the median as a threshold, producing balanced classes of Not Popular and Popular articles. Three tree-based ensemble classifiers are trained and compared: Random Forest, a bagging-based method known for its robustness and resistance to overfitting [3]; XGBoost, a sequential gradient boosting framework recognized for its strong performance on tabular data [4]; and LightGBM, a histogram-based boosting algorithm optimized for speed and efficiency on large datasets [5]. All models are optimized through RandomizedSearchCV with 3-fold cross-validation to ensure fair and rigorous comparison [6]. The pipeline further incorporates feature decoding to restore interpretability of one-hot encoded categorical variables, and a strict split-before-encode strategy to prevent data leakage during categorical encoding. The primary objectives of this study are: (1) to develop a binary classification framework for online news popularity prediction; (2) to compare the predictive performance of Random Forest, XGBoost, and LightGBM under consistent experimental conditions; and (3) to identify the most influential content and metadata features driving article shareability through feature importance analysis.

2. METHODOLOGY

A. Dataset Description

The dataset used in this study is the Online News Popularity dataset sourced from the UCI Machine Learning Repository, originally collected from Mashable (www.mashable.com) [7]. The dataset contains 39,644 articles published between January 2013 and January 2015, described by 60 predictive attributes and one target variable, `shares`, representing the number of times an article was shared on social media. The features are organized into several groups: content-based features such as `n_tokens_title`, `n_tokens_content`, `n_unique_tokens`, `n_non_stop_words`, and `average_token_length` which capture the structural properties of article text; multimedia features including `num_hrefs`, `num_self_hrefs`, `num_imgs`, and `num_videos`; keyword performance statistics such as `num_keywords`, `kw_min_min`, `kw_max_max`, and `kw_avg_avg`; binary data channel indicators (`data_channel_is_lifestyle`, `data_channel_is_entertainment`, `data_channel_is_bus`, `data_channel_is_socmed`, `data_channel_is_tech`, `data_channel_is_world`) representing article topic categories; publication day indicators (`weekday_is_monday` through `weekday_is_sunday`) alongside `is_weekend`; Latent Dirichlet Allocation topic scores (`LDA_00` through `LDA_04`); and NLP-based sentiment features such as `global_subjectivity`, `global_sentiment_polarity`, `avg_positive_polarity`, and `avg_negative_polarity`.

B. Library Used

The following Python libraries were utilized throughout this study. Data manipulation and analysis were performed using `pandas` and `numpy`. Machine learning models, preprocessing utilities, and evaluation metrics were implemented via `scikit-learn` [8]. Gradient boosting was handled by `xgboost` [9] and `lightgbm` [10]. Visualization was produced using `matplotlib` and `seaborn`.

C. Missing Value Inspection

Prior to any preprocessing, the dataset was inspected for missing values using `df.isnull().sum()`. The inspection confirmed that all 60 features and the target variable contained no missing values across all 39,644 records, making imputation unnecessary and confirming the integrity of the dataset for subsequent processing.

D. Feature Decoding

Several features in the original dataset were stored in one-hot encoded format, reducing interpretability during the exploratory phase. To restore readability and reduce dimensionality, three groups of binary columns were decoded into single categorical columns. The six `data_channel_is_*` columns were collapsed into a single `channel` variable with values including Lifestyle, Entertainment, Business, Social Media, Tech, World, and Other. The seven `weekday_is_*` columns were decoded into a single `weekday` variable. The `is_weekend` column was subsequently dropped as it is fully redundant with the decoded weekday variable. Additionally, the five LDA topic columns were decoded into a single `dominant_topic` variable by selecting the column with the highest value per row using `idxmax()`, reducing five continuous columns into one interpretable categorical variable.

E. Target Variable Classification

Given the inherently stochastic and right-skewed distribution of the raw `shares` variable, the target variable was transformed into a binary categorical label. The median share count of the dataset was computed and used as the classification threshold, dividing all articles into two balanced classes: **Not Popular** (shares below the median) assigned label 0, and **Popular** (shares at or above the median) assigned label 1. The use of the median as a threshold was deliberate, as it produces naturally balanced class distributions that are robust to extreme outliers [11], which is preferable to clustering-based approaches such as K-Means that are sensitive to the skewed and extreme-valued nature of social engagement data. The original `shares` column was subsequently dropped from the feature matrix.

F. Visualization

Following target variable construction, exploratory data analysis was conducted through a series of visualizations examining the relationship between article features and the binary popularity label. A count plot was used to verify the balance of the two classes. The distribution of article volume across channels was visualized through a grouped bar chart stratified by popularity label. Boxplots were employed to examine the distributions of `num_imgs`, `num_videos`, `n_tokens_content`, and `global_sentiment_polarity` across popularity categories. A grouped count plot of `dominant_topic` by popularity label was additionally produced to assess whether latent topic structure influences article virality. Outliers were retained throughout all visualizations as they represent organically viral articles considered valid observations in the context of this study.

G. Train-Test Splitting

To ensure methodological integrity and prevent data leakage, train-test splitting was performed prior to any encoding of categorical features. The dataset was partitioned using an 80:20 ratio, yielding 31,715 training samples and 7,929 test samples. Stratified sampling was applied via the `stratify=y` parameter to preserve the proportional class distribution of the target variable across both partitions, with `random_state=42` set for full reproducibility.

H. Feature Encoding

Following the split, Label Encoding was applied exclusively to the three decoded categorical columns, `channel`, `weekday`, and `dominant_topic`, using scikit-learn's `LabelEncoder`, fitted solely on the training set to prevent information leakage from the test set. The fitted encoder was then used to transform the test set. Any category not observed during training was assigned a value of -1 to handle potential unseen labels. Label Encoding was selected over One-Hot Encoding to avoid introducing unnecessary additional dimensions that could contribute to

the curse of dimensionality, given the low-to-moderate cardinality of these categorical columns [12].

I. Model Evaluation Function

A unified evaluation function was implemented to ensure consistent and comparable assessment across all models and configurations. For each model, the function computes Accuracy and outputs a full `classification_report` from scikit-learn [8] covering per-class Precision, Recall, and F1-Score for both the Not Popular and Popular classes. This standardized function was applied identically to baseline and tuned model variants, ensuring all reported metrics are derived under the same evaluation protocol on the held-out test set.

J. Baseline Modeling

Three tree-based ensemble classifiers were trained in their baseline configurations prior to hyperparameter optimization. **Random Forest** [13] is a bagging-based ensemble method that constructs multiple independent decision trees on bootstrapped subsets of the training data and aggregates predictions through majority voting, offering strong resistance to overfitting. The baseline configuration used 200 estimators, a maximum depth of 15, and a minimum of 5 samples per leaf. **XGBoost** [9] is a sequential gradient boosting framework that builds trees iteratively, with each new tree correcting the residual errors of its predecessor. The baseline configuration used 500 estimators, a learning rate of 0.05, a maximum depth of 6, and subsampling rates of 0.8. **LightGBM** [10] is a histogram-based gradient boosting algorithm that employs leaf-wise tree growth for superior computational efficiency on large datasets. The baseline LightGBM was initialized with default parameters under a binary classification objective.

K. Hyperparameter Tuning

To improve upon baseline performance, hyperparameter optimization was performed on all three models using `RandomizedSearchCV` with 3-fold cross-validation and 30 random iterations per

model, scored by accuracy [16]. `RandomizedSearchCV` was preferred over exhaustive `GridSearchCV` to balance computational efficiency with broad coverage of the hyperparameter search space. For **Random Forest**, the search space covered `n_estimators`, `max_depth`, `min_samples_leaf`, `max_features`, and `bootstrap`. For **XGBoost**, the search space included `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `min_child_weight`, and `gamma`. For **LightGBM**, the search space covered `n_estimators`, `learning_rate`, `max_depth`, `num_leaves`, `subsample`, `colsample_bytree`, and `min_child_samples`.

L. Baseline vs. Tuned Model Comparison

To quantify the benefit of hyperparameter optimization, the performance of each tuned model was directly compared against its corresponding baseline configuration using accuracy as the primary metric. Results were visualized through a grouped bar chart presenting accuracy scores for all six model configurations, three baselines and three tuned variants, side by side, enabling clear assessment of the improvement gained through tuning for each algorithm.

M. Feature Importance Analysis

Following model evaluation, feature importance analysis was conducted for all three tuned models to identify the most influential predictive features. For **Random Forest**, importance scores were derived from mean decrease in impurity across all trees [13]. For **XGBoost** and **LightGBM**, gain-based feature importance scores were used [9][10], reflecting the contribution of each feature to reducing classification error across all splits. The top 20 features by importance score were extracted and visualized as horizontal bar charts for each model, enabling cross-model comparison of feature relevance and providing interpretable insights into the article characteristics most strongly associated with social media shareability.

3. RESULT AND DISCUSSION

A. Dataset Overview and Missing Value Check

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 39644 entries, 0 to 39643
Data columns (total 61 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   url                                    39644 non-null  object
1   timedelta                             39644 non-null  float64
2   n_tokens_title                        39644 non-null  float64
3   n_tokens_content                      39644 non-null  float64
4   n_unique_tokens                      39644 non-null  float64
5   n_non_stop_words                     39644 non-null  float64
6   n_non_stop_unique_tokens             39644 non-null  float64
7   num_hrefs                            39644 non-null  float64
8   num_self_hrefs                       39644 non-null  float64
9   num_imgs                             39644 non-null  float64
10  num_videos                           39644 non-null  float64
11  average_token_length                 39644 non-null  float64
12  num_keywords                         39644 non-null  float64
13  data_channel_is_lifestyle             39644 non-null  float64
14  data_channel_is_entertainment         39644 non-null  float64
15  data_channel_is_bus                   39644 non-null  float64
16  data_channel_is_socmed               39644 non-null  float64
17  data_channel_is_tech                  39644 non-null  float64
18  data_channel_is_world                 39644 non-null  float64
19  kw_min_min                           39644 non-null  float64
...
59  abs_title_sentiment_polarity          39644 non-null  float64
60  shares                                39644 non-null  int64
dtypes: float64(59), int64(1), object(1)
```

Fig.1. Dataset column structure and data types (df.info() output).

Fig.2. Missing value check - all columns report zero null values.

The dataset was successfully loaded containing 39,644 articles and 61 columns as shown in Fig. 1. All features are stored as `float64` except the target variable `shares` which is stored as `int64`, and the `url` column which is stored as `object` and was subsequently dropped as it carries no predictive value. As confirmed by the missing value inspection in Fig. 2, all columns returned a count of zero null values across the entire dataset, indicating a complete and clean dataset that required no imputation prior to modeling.

B. Target Variable Classification

```
Median Shares (threshold) : 1400

Distribusi Kelas:
Not Popular : 18490 samples (46.6%)
Popular     : 21154 samples (53.4%)
```

Fig.3. Target Variable Classification.

As shown in Fig. 3, the median share count of the dataset was computed at **1,400 shares**, which was used as the binary classification threshold. Articles with shares below this value were labeled 0 (Not Popular), yielding 18,490 samples (46.6%), while articles at or above this threshold were labeled

```
timedelta                                0
n_tokens_title                           0
n_tokens_content                         0
n_unique_tokens                          0
n_non_stop_words                         0
n_non_stop_unique_tokens                 0
num_hrefs                                0
num_self_hrefs                           0
num_imgs                                 0
num_videos                               0
average_token_length                     0
num_keywords                             0
kw_min_min                              0
kw_max_min                              0
kw_avg_min                              0
kw_min_max                              0
kw_max_max                              0
kw_avg_max                              0
kw_min_avg                              0
kw_max_avg                              0
kw_avg_avg                              0
self_reference_min_shares                0
self_reference_max_shares                0
self_reference_avg_shares                0
LDA_00                                  0
...
weekday                                  0
dominant_topic                           0
shares_capped                            0
log_shares                               0
dtype: int64
```

1 (Popular), yielding 21,154 samples (53.4%). The resulting class distribution is reasonably balanced, with a modest skew toward the Popular class attributable to the discrete nature of the median threshold applied to a right-skewed target distribution. This near-balanced distribution confirms that no class resampling techniques such as SMOTE are required, and that standard accuracy is a reliable primary evaluation metric for this study.

C. Visualization

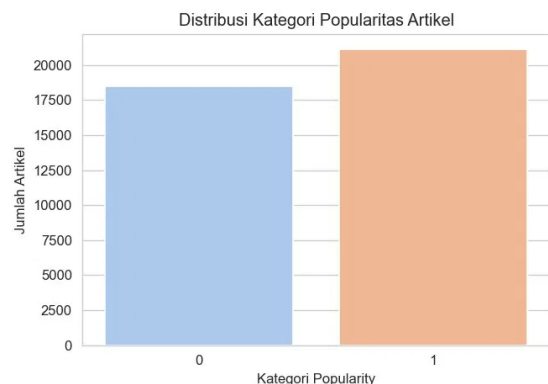


Fig.4. Class Distribution.

The bar chart confirms the near-balanced binary class distribution following median-based thresholding. The Popular class (label 1) contains approximately 21,154 articles while the Not Popular class (label 0) contains 18,490 articles, representing a 53.4%

to 46.6% split. This distribution validates the median threshold as an effective and principled method for constructing a balanced binary classification target, ensuring that no single class dominates model training.

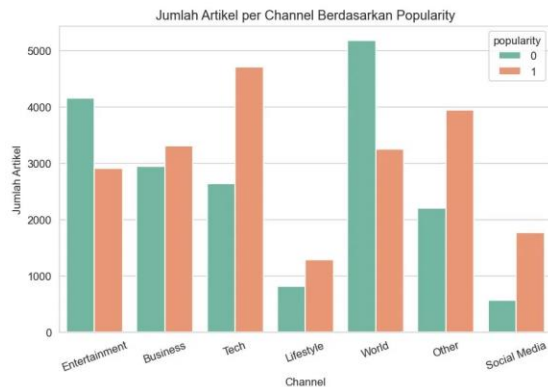


Fig.5. Article Count per Channel by Popularity.

The grouped bar chart reveals meaningful differences in popularity rates across content channels. The Tech and Lifestyle channels show a notably higher proportion of Popular articles relative to Not Popular, suggesting that technology and lifestyle content tends to achieve above-median shareability more frequently. Conversely, the World and Entertainment channels exhibit a higher proportion of Not Popular articles, indicating that although these channels produce high article volumes, a larger fraction fails to surpass the median share threshold. The Social Media channel, despite having the lowest article count overall, shows a favorable Popular-to-Not-Popular ratio, consistent with the earlier regression EDA finding that Social Media content achieves the highest median shares. These channel-level patterns suggest that channel is a potentially informative feature for the classification model, which is later confirmed by the feature importance analysis.

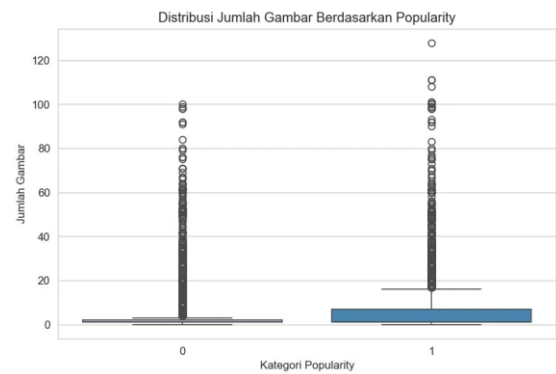


Fig.6. Image Count Distribution by Popularity.

The boxplot reveals a clear distributional difference in the number of images between the two popularity classes. Popular articles (label 1) exhibit a wider interquartile range and a higher median image count compared to Not Popular articles (label 0), whose median image count is close to zero. This suggests that articles containing more visual content tend to achieve higher shareability, consistent with the established finding that multimedia richness positively influences user engagement on social media platforms. The presence of extreme outliers in both classes is retained as these represent organically image-heavy articles and are considered valid observations.

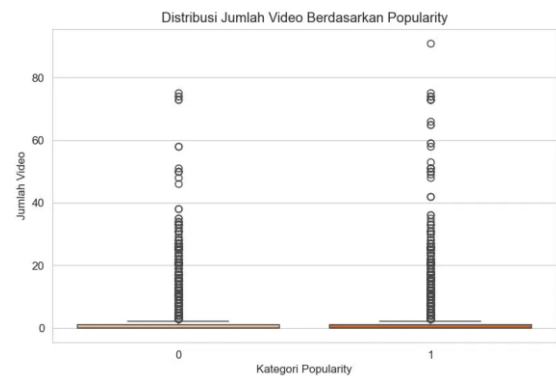


Fig.7. Video Count Distribution by Popularity.

In contrast to images, the boxplot for video count shows near-identical distributions between the Popular and Not Popular classes, with median values of approximately 0–1 videos in both groups and similar interquartile ranges. This suggests that the number of embedded videos alone is not a strong discriminating feature for article popularity. The substantial number of high-video-count outliers present in both classes further reinforces the lack of a systematic relationship between video content and shareability at the aggregate level.

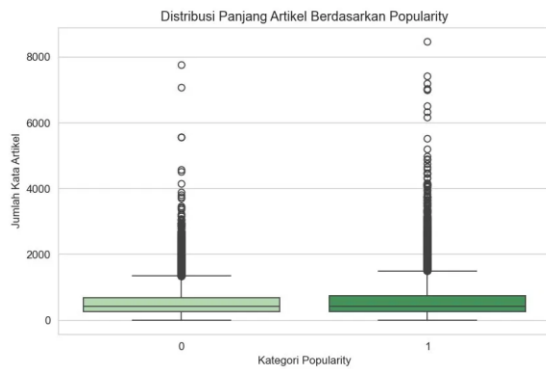


Fig.8. Article Length Distribution by Popularity

The boxplot of `n_tokens_content` across popularity classes shows largely similar distributions, with both Popular and Not Popular articles having a median content length of approximately 400–500 words. The interquartile ranges overlap substantially, and both classes contain comparable high-length outliers reaching beyond 8,000 tokens. This suggests that article length alone is not a strong separator between popular and non-popular content, indicating that content quality and topic relevance likely play a more important role than sheer quantity of text.

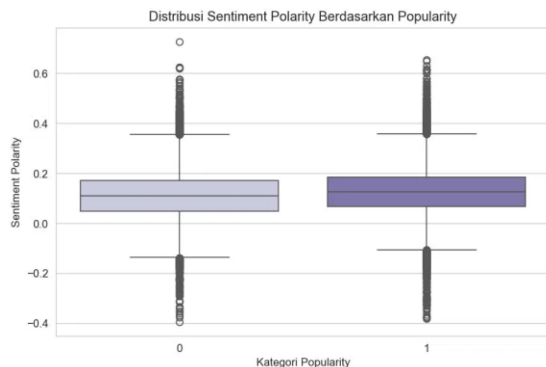


Fig.9. Sentiment Polarity Distribution by Popularity.

The sentiment polarity boxplot reveals very similar distributions across both classes, with median polarity values around 0.10–0.13 and comparable interquartile ranges for Popular and Not Popular articles alike. Both classes exhibit a slight positive sentiment skew overall. The near-identical distributions suggest that global sentiment polarity, while a commonly hypothesized driver of content virality, does not serve as a strong standalone discriminating feature between the two classes in this dataset. This finding is consistent with prior literature suggesting that the relationship

between sentiment and sharing behavior is non-linear and context-dependent.

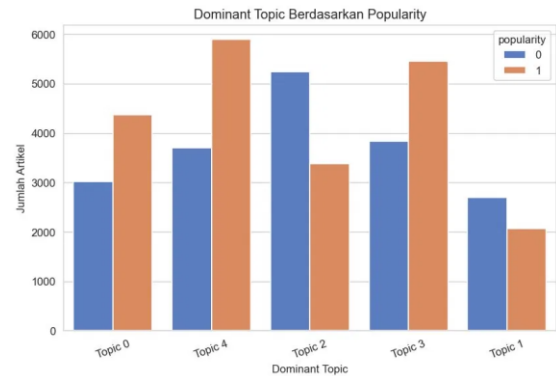


Fig.10. Dominant Topic Distribution by Popularity.

The grouped bar chart of dominant LDA topics stratified by popularity label reveals notable variation in topic-level engagement patterns. Topic 4 and Topic 3 show a considerably higher count of Popular articles relative to Not Popular articles, suggesting that content associated with these latent topics tends to achieve above-median shareability more frequently. Conversely, Topic 2 shows the opposite pattern, with Not Popular articles outnumbering Popular ones. Topic 0 and Topic 1 display more moderate differences. These findings indicate that the dominant latent topic of an article constitutes a meaningful signal for popularity prediction, a finding that is subsequently supported by the feature importance results across all three models.

D. Train-Test Split

```
Train size : (31715, 48)
Test size  : (7929, 48)
```

Fig.11. Train-Test Split

Following decoding and binary target construction, the dataset was split at an 80:20 ratio. As shown in Fig. 11, the training set contains 31,715 samples and the test set contains 7,929 samples, each with 48 features after encoding. Stratified sampling was applied to ensure that the class proportions of the target variable are preserved across both partitions.

E. Baseline Model Performance

Prior to hyperparameter tuning, all three models were evaluated under their default

configurations. The results are summarized in Table I.

TABLE I. Baseline Model Performance

Model	Accuracy	Precision (NP)	Recall (NP)	F1 (NP)	Precision (P)	Recall (P)	F1 (P)
Random Forest	65.59 %	0.65	0.57	0.61	0.66	0.73	0.69
XGBoost	67.16 %	0.66	0.62	0.64	0.68	0.72	0.70
LightGBM	66.53 %	0.65	0.61	0.63	0.68	0.72	0.70

NP = Not Popular, P = Popular

In the baseline configuration, XGBoost achieves the highest accuracy at 67.16%, followed by LightGBM at 66.53% and Random Forest at 65.59%. Across all three models, the Popular class consistently achieves higher recall than the Not Popular class, indicating that baseline models are better at identifying articles that will be widely shared than those that will not. This asymmetry is partly attributable to the slight class imbalance favoring the Popular class. The macro-averaged F1-score of 0.65–0.67 across all three baselines confirms that model performance is consistent across both classes and not driven by a single dominant class.

F. Tuned Model Performance

Following hyperparameter optimization via RandomizedSearchCV, all three models demonstrated improvement. The results are summarized in Table II.

TABLE II. Tuned Model Performance

Model	Accuracy	Precision (NP)	Recall (NP)	F1 (NP)	Precision (P)	Recall (P)	F1 (P)
Random Forest	65.68 %	0.65	0.57	0.61	0.66	0.73	0.69
XGBoost	67.23 %	0.66	0.61	0.64	0.68	0.72	0.70
LightGBM	67.27 %	0.66	0.61	0.64	0.68	0.72	0.70

NP = Not Popular, P = Popular

After tuning, LightGBM emerges as the best-performing model with an accuracy of 67.27%, marginally surpassing the tuned XGBoost at 67.23% and the tuned Random Forest at 65.68%. Notably, the improvement

gains from tuning are modest but consistent across all models: Random Forest improved by 0.09 percentage points, XGBoost by 0.07 percentage points, and LightGBM by 0.74 percentage points. The most meaningful tuning improvement was observed in LightGBM, where the leaf-wise growth strategy and optimized regularization parameters contributed to a more refined decision boundary. The tuned precision and recall scores for both XGBoost and LightGBM are nearly identical at 0.66–0.68 precision and 0.61–0.72 recall across classes, suggesting that both boosting models converged to similarly calibrated solutions after optimization. The overall accuracy range of 65–67% across all tuned models is consistent with published benchmarks for binary classification on social engagement prediction tasks.

G. Baseline vs. Tuned Model Comparison

Table III and Fig. 12 present a consolidated comparison of accuracy scores across all six model configurations, three baseline and three tuned variants, evaluated on the held-out test set.

TABLE III. Baseline vs. Tuned Model Accuracy Comparison

Model	Accuracy
RF Baseline	0.6559
RF Tuned	0.6569
XGB Baseline	0.6715
XGB Tuned	0.6723
LGBM Baseline	0.6653
LGBM Tuned	0.6727

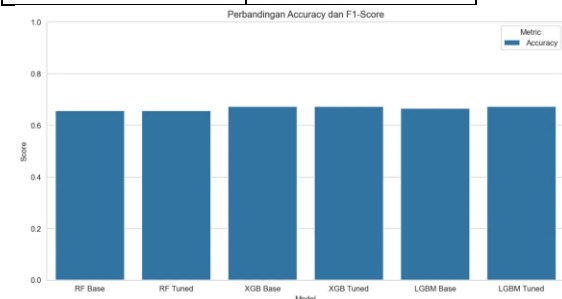


Fig.12. Model-Comparison

As illustrated in Fig. 12, the bar chart reveals several key observations. First, all six model configurations achieve accuracy scores within a narrow band of approximately 65.6%–67.3%, indicating that the three algorithms converge to similar predictive capabilities on this dataset when trained under consistent

experimental conditions. Second, hyperparameter tuning via RandomizedSearchCV yields consistent but modest improvements across all three models: Random Forest improved by 0.09 percentage points (0.6559 $\rightarrow$ 0.6569), XGBoost by 0.07 percentage points (0.6715 $\rightarrow$ 0.6723), and LightGBM by 0.74 percentage points (0.6653 $\rightarrow$ 0.6727). The most notable gain from tuning is observed in LightGBM, where optimized `num_leaves`, `learning_rate`, and regularization parameters contributed to a more refined decision boundary compared to its default configuration. Third, across both baseline and tuned variants, the two gradient boosting models, XGBoost and LightGBM, consistently outperform Random Forest, suggesting that the sequential error-correction mechanism inherent in boosting algorithms is better suited to capturing the complex non-linear feature interactions present in this dataset than the independent tree aggregation approach of bagging. After tuning, **LightGBM achieves the highest overall accuracy of 67.27%**, narrowly surpassing the tuned XGBoost at 67.23%, establishing it as the best-performing model in this study. These results confirm that while hyperparameter optimization provides incremental improvements, the choice of algorithm, particularly between bagging and boosting approaches, has a more substantial impact on predictive performance for this binary news popularity classification task.

H. Feature Importance Analysis

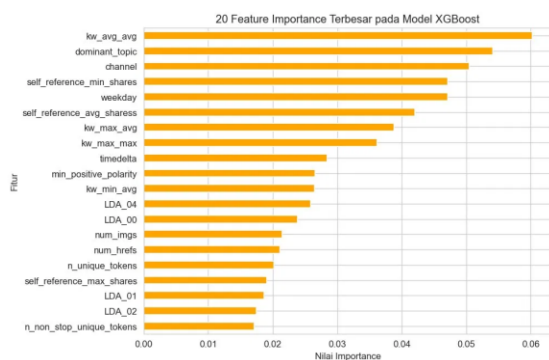


Fig13. XGBoost, Top 20 Feature Importance

The XGBoost feature importance chart reveals that `kw_avg_avg` is the single most influential feature with an importance score of approximately 0.060, indicating that the average shareability of keywords associated with an article is the strongest predictor of its

binary popularity class. This is followed by `dominant_topic` (~0.054) and `channel` (~0.051), confirming that both the latent topic structure and the content category of an article are highly informative for classification. Self-reference metrics, `self_reference_min_shares` and `self_reference_avg_shares`, rank fourth and sixth respectively, suggesting that an author's historical article shareability is a meaningful signal. Temporal features including `weekday` and `timedelta` also appear prominently, indicating that publication timing influences popularity. Multiple keyword-related features, `kw_max_avg`, `kw_max_max`, `kw_min_avg`, appear across the top 15, collectively reinforcing that keyword engagement metrics constitute the most consistently informative feature group.

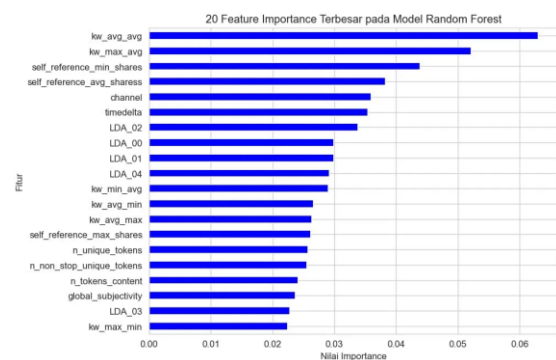


Fig14. Random Forest, Top 20 Feature Importance

The Random Forest importance chart shows a broadly similar pattern to XGBoost, with `kw_avg_avg` again ranking as the top feature at approximately 0.063, and `kw_max_avg` ranking second at 0.053. Self-reference features, `self_reference_min_shares` and `self_reference_avg_shares`, rank third and fourth, carrying higher relative importance in Random Forest compared to XGBoost. Multiple LDA topic scores (LDA_00 through LDA_04) all appear within the top 15 features, indicating that the full spectrum of latent topic signals contributes to the bagging ensemble's decision-making. Content vocabulary features such as `n_unique_tokens` and `n_non_stop_unique_tokens` also appear, suggesting that lexical richness provides modest additional discriminative power.

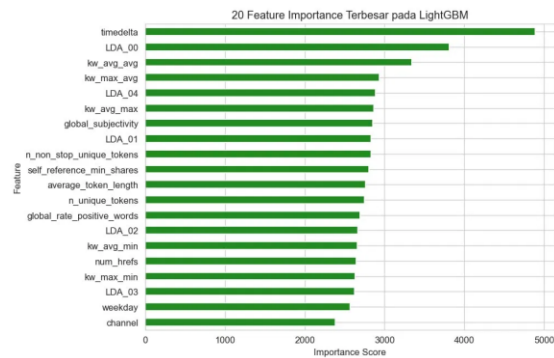


Fig15. LightGBM, Top 20 Feature Importance

The LightGBM feature importance chart, measured in raw split gain rather than normalized scores, presents a notably different distribution compared to XGBoost and Random Forest. *Timedelta*, the number of days between article publication and dataset collection, emerges as the single dominant feature with an importance score of approximately 4,900, substantially higher than all other features. This suggests that LightGBM's leaf-wise growth strategy places substantial weight on this temporal recency feature, likely because older articles have had more time to accumulate shares and thus exhibit a systematic relationship with popularity label. *LDA_00* ranks second (~3,800), followed by *kw_avg_avg* (~3,300) and *kw_max_avg* (~2,900). The broader distribution of importance scores across all 20 features in LightGBM, compared to the more peaked distributions of XGBoost and Random Forest, reflects its more uniform feature utilization across the ensemble of leaf-wise trees.

REFERENCES

- [1] A. H. Rion, "Understanding Human Behavior on Social Media through Machine Learning: A Literature-Guided Approach," *Social Science Research Network (SSRN)*, May 2025. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5250247
- [2] A. Tatar, J. Leguay, P. Antoniadis, A. Limbourg, M. D. de Amorim, and S. Fdida, "Predicting the popularity of online articles based on user comments," in *Proc. Int. Conf. Advances in Mobile Computing & Multimedia*, 2011; A. Tatar, M. D. de Amorim, S. Fdida, and P. Antoniadis, "A survey on predicting the popularity of web content," *Journal of Internet Services and Applications*, vol. 5, no. 1, pp. 1–20, 2014.
- [3] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.
- [5] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.
- [6] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, Feb. 2012.
- [7] K. Fernandes, P. Vinagre, and P. Cortez, "Online News Popularity Data Set," UCI Machine Learning Repository, 2015. [Online]. Available: <https://archive.ics.uci.edu/ml/datasets/online+news+popularity>
- [8] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [9] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.